Pengcheng You

Xin'ao Engineering Building 2045, Peking University,
5 Yiheyuan Rd, Haidian District, Beijing, China, 100871
https://pengcheng-you.github.io/desires-lab/
pcyou@pku.edu.cn +86 (10) 6276 8229

EDUCATION

Ph.D., Control Science and Engineering
Zhejiang University, Hangzhou, China *Sep. 2013 - Jun. 2018*

B.S. with honors, Electrical Engineering
Zhejiang University, Hangzhou, China *Sep. 2009 - Jun. 2013*

PRIMARY APPOINTMENTS

Assistant Professor, Dept. of Control Science and Systems Engineering
Peking University, Beijing, China *Jul. 2025 - Present*

Assistant Professor, State Key Laboratory for Turbulence and Complex Systems
Peking University, Beijing, China *Apr. 2025 - Present*

Assistant Professor, National Engineering Laboratory for Big Data Analysis and Applications
Peking University, Beijing, China *Dec. 2022 - Present*

Assistant Professor, Dept. of Industrial Engineering and Management
Peking University, Beijing, China *Feb. 2022 - Jun. 2025*

Lecturer, Engineering for Professionals Program
Johns Hopkins University, Baltimore, MD, US *Aug. 2021 - Dec. 2021*

Postdoctoral Fellow, Depts. of Mechanical Engineering/Electrical and Computer Engineering
Johns Hopkins University, Baltimore, MD, US *Oct. 2018 - Dec. 2021*

Research Intern, Data Sciences and Analytics Group
Pacific Northwest National Laboratory, Richland, WA, US *Oct. 2017 - Jan. 2018*

Visiting Graduate Student, Dept. of Computing and Mathematical Sciences
California Institute of Technology, Pasadena, CA, US *Oct. 2016 - Oct. 2017*

AWARDS

- Boya Young Fellow - Peking University *2025*
- Excellent Teaching Team Award - Peking University (Engineering Data Analysis and Optimization Teaching Team) *2024*
- Overseas Talent - China National Administrative Committee of Post-doctoral Researchers *2024*
- Startrack Scholar - Microsoft Research Aisa *2023*
- ACM e-Energy Student Travel Grant (10 awardees) *2018*
- Sifang Scholarship - Zhejiang University *2017, 2016*
- Outstanding Graduate Student - Zhejiang University *2017, 2016, 2015*
- Chu Kochen Honors Bachelor Degree - Zhejiang University (∼5% of graduates) *2013*
- Excellent Student Scholarship - Chu Kochen Honors College, Zhejiang University *2012, 2010*

Students' awards

- ACC'26 Student Travel Award - Yixuan Yu *2026*
- CDC'25 Student Travel Award - Chengrui Qu *2025*

- PKU Outstanding Graduate - Chengrui Qu *2025*
- PKU Students of the Year Finalist (Final 20) - Zhuo Ouyang *2024*
- National Scholarship - Zhuo Ouyang *2024*
- National College Student Innovation & Entrepreneurship Development Program - Zijian Zhao *2024*
- High-Quality Paper Award, Industrial Engineering Committee Annual Meeting of Chinese Society of Optimization, Overall Planning and Economic Mathematics - Zijian Zhao *2024*
- Beijing Municipal NSF Qiyan Program for Undergraduates - Zhuo Ouyang *2024*
- DTU PES Summer School Best Poster Award - Chengrui Qu *2024*
- CDC'24 Student Travel Award and Workshop Support - Yingzhu Liu *2024*
- Caltech Summer Undergraduate Research Fellowship (SURF) - Chengrui Qu *2024*
- National Scholarship - Chengrui Qu *2023*
- Dean's First-Class Scholarship - Kekun Gao *2023*
- NSFC 1st Youth Student Basic Research Grant - Chengrui Qu *2023*

TEACHING

Instructor

- Foundations of Reinforcement Learning *Peking University, Spring 2026*
- Probability and Statistics *Peking University, Fall 2025*
- Advanced Operations Research *Peking University, Spring 2025*
- Probability and Statistics *Peking University, Fall 2024*
- Foundations of Reinforcement Learning *Peking University, Spring 2024*
- Probability and Statistics *Peking University, Fall 2023*
- Advanced Operations Research *Peking University, Spring 2023*
- Foundations of Reinforcement Learning *Peking University, Fall 2022*
- Foundations of Reinforcement Learning *Johns Hopkins University, Fall 2021*

Teaching assistant

- Energy System Analysis *Johns Hopkins University, Spring 2021*
- Analytical Methods for Network Congestion Control *Zhejiang University, Summer 2016*
- Power Systems Analysis *Zhejiang University, Summer 2015*
- Linear Systems Analysis *Zhejiang University, Fall 2013*

Teaching reform project

- Probability and Statistics - College of Engineering, Peking University *2024*
  – *Outstanding Completed Project*

PROJECTS

Grants awarded

1. **State Grid**: *Data-Driven Methods for Solving Power Flows: Key Techniques and Pilot Application*, 2026-2027
2. **NRF of Korea**: *A Study on AI Agent for Product Development and Organizational Integration*, 2025-2028
3. **NSFC**: *Multi-Scale Optimization with Uncertainty and Its Application in Industrial Park Energy System Optimization*, 2025-2029
4. **PKU Office of International Relations**: *Advancing Sustainable Transportation: Battery Degradation-Aware Electric Vehicle Charging for Net Zero*, 2024
5. **Overseas Talent Program**: *Enabling Integration of Distributed Energy Resources via Redesign of Power Grid Information and Management Architecture*, 2024-2026
6. **State Grid Research Project**: *Trustworthy AI for Power System Dispatch*, 2024-2025

7. **Ali Cloud & Damo Academy Research Project**: *Carbon Emission Constrained Optimization for Multi-Scale Economic Operation and Electricity Trading Strategy of Data Centers*, 2023-2024
8. **166 Research Project**: *Machine Learning Enabled Algorithms for Complex Strategic Resource Scheduling*, 2022-2024
9. **Huawei Cloud Research Project**: *Modeling, Simulation, Evaluation, and Causal Analysis of Resource Management Platform for Huawei Public Cloud IaaS Service*, 2022-2023
10. **NSFC** (72201007): *Digital Twin Driven Management for Electric Vehicle Charging: Evaluation and Realization of Its Economic Values*, 2023-2025
11. **PKU Emerging Engineering Interdisciplinary Project**: *Incentive Analysis for Coupled Participation in Electricity and Carbon Markets*, 2022
12. **PKU College of Engineering Startup Fund**: *Data-Driven Modeling, Analysis and Design for Electricity Markets*, 2022-2024

Grants as main contributor

1. **NSF CPS Medium** (2136324): *Enabling DER Integration via Redesign of Information Flows*, 2021-2024
2. **NSFC** (61973163): *Countermeasures for Deception Attack on Networked Control Systems*, 2020-2023
3. **JHU Discovery Award**: *A Data-Driven Approach to Electricity Market Design*, 2019-2021
4. **NSF** (1711188): *An Optimization Decomposition Framework for Principled Multi-Timescale Market Design and Co-Optimization*, 2017-2020
5. **DOE** (DE-EE0008006): *Robust and resilient coordination of feeders with uncertain distributed energy resources: from real-time control to long-term planning*, 2017-2020

FULL PUBLICATIONS

(Underlined include students or postdocs advised.)

Book chapters

1. P. You. **Scheduling of EV Battery Swapping in Microgrid**. Book chapter in *Distributed Control Methods and Cyber Security Issues in Microgrids* (Editors: X. Wang, W. Meng, S. Liu; ISBN: 978-0-12-816946-9), pp. 203-241, Elsevier Academic Press, 2020

Journals

1. R. Chen, P. You, Z. Yang, and K. Tang. **Multi-Vehicle Cooperative Motion Planning based on ADMM with Parallel Computing and Convexifying Guidance**. *IEEE Transactions on Intelligent Transportation Systems*, accepted
2. R. K. Bansal, Y. Chen, P. You, and E. Mallada. **Counterfactual Analysis of Default Bid Market Power Mitigation Strategies in Two-Stage Electricity Markets**. *European Journal of Operational Research*, accepted
3. K. Gao*, Y. Yan*, Y. Zhou*, E. Liu, Z. Wang, and P. You. **Optimal Renewable Power Purchase Agreements for Data Centers**. *IEEE Transactions on Smart Grid*, accepted (*equal contributions)
4. Z. Zhao, Y. Qiu, and P. You. **From Ambiguity to Transparency: Blockchain-Enabled Origin Traceability for Premium Agricultural Product**. *Transportation Research Part E*, accepted
5. P. You, Y. Jiang, E. Yeung, D. F. Gayme, and E. Mallada. **On the Stability, Economic Efficiency and Incentive Compatibility of Electricity Market Dynamics**. *IEEE Transactions on Automatic Control*, accepted
6. T. Zheng, N. Loizou, P. You, and E. Mallada. **Dissipative Gradient Descent Ascent Method: A Control Theory Inspired Algorithm for Min-max Optimization**. *IEEE Control Systems Letters (L-CSS)*, vol. 8, pp. 2009-2014, Jun. 2024
7. L. Mo, X. Li, A. Kritikakou, and P. You. **Optimal IC Task Mapping to Maximize QoS on Heterogeneous Multicore Systems**. *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 71, no. 4, pp. 2019-2023, Apr. 2024

8. R. Chen, Z. Liang, J. Cheng, P. You, and Z. Yang. **Multi-Agent Cooperative Motion Planning based on Alternating Direction Method of Multipliers**. *IEEE Control Systems Letters* (*L-CSS*), vol. 7, pp. 3307-3312, Oct. 2023
9. R. K. Bansal, Y. Chen, P. You, and E. Mallada. **Market Power Mitigation in Two-stage Electricity Market with Supply Function and Quantity Bidding**. *IEEE Transactions on Energy Markets, Policy and Regulation*, vol. 1, no. 4, pp. 512-522, Dec. 2023
10. R. K. Bansal, P. You, D. F. Gayme, and E. Mallada. **A Market Mechanism for Truthful Bidding with Energy Storage**. *Electric Power Systems Research*, vol. 211, 108284, Oct. 2022
11. R. Pates, A. Ferragut, E. Pivo, P. You, F. Paganini, and E. Mallada. **Respect the Unstable: Delays and Saturation in Contact Tracing for Disease Control**. *SIAM Journal on Control and Optimization*, vol. 60, no. 2, pp. S196-S220, Mar. 2022
12. J. Sun, S. Chen, P. You, Q. Yang, and Z. Yang. **Battery-Assisted Online Operation of Distributed Data Centers with Uncertain Workload and Electricity Prices**. *IEEE Transactions on Cloud Computing*, vol. 11, no. 2, pp. 1303-1314, Jun. 2023
13. M. Almassalkhi, *et al*. **Hierarchical, Grid-Aware, and Economically Optimal Coordination of Distributed Energy Resources in Realistic Distribution Systems**. *Energies*, vol. 13, no. 23, 6399, Dec. 2020
14. H. Yang, P. You, and C. Shang. **Distributed Planning of Electricity and Natural Gas Networks and Energy Hubs**. *Applied Energy*, vol. 282, Part A, 116090, Jan. 2021
15. P. You, J. Z.F. Pang, and S. H. Low. **Online Station Assignment for Electric Vehicle Battery Swapping**. *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 4, pp. 3256-3267, Apr. 2022
16. L. Mo, P. You, X. Cao, and Y.-Q. Song. **Event-Driven Joint Mobile Actuators Scheduling and Control in Cyber-Physical Systems**. *IEEE Transactions on Industrial Informatics*, vol. 15, no. 11, pp. 5877-5891, Nov. 2019
17. Z. Yang, T. Guo, P. You, Y. Hou, and S. J. Qin. **Distributed Approach for Temporal-Spatial Charging Coordination of Plug-in Electric Taxi Fleet**. *IEEE Transactions on Industrial Informatics*, vol. 15, no. 6, pp. 3185-3195, Jun. 2019
18. Y. Zhang, P. You, and L. Cai. **Optimal Charging Scheduling by Pricing for EV Charging Station with Dual Charging Modes**. *IEEE Transactions on Intelligent Transportation Systems*, vol. 20, no. 9, pp. 3386-3396, Sep. 2019
19. P. You, S. H. Low, L. Zhang, R. Deng, G. B. Giannakis, Y. Sun, and Z. Yang. **Scheduling of EV Battery Swapping, II: Distributed Solutions**. *IEEE Transactions on Control of Network Systems*, vol. 5, no. 4, pp. 1920-1930, Dec. 2018
20. P. You, S. H. Low, W. Tushar, G. Geng, C. Yuen, Z. Yang, and Y. Sun. **Scheduling of EV Battery Swapping, I: Centralized Solution**. *IEEE Transactions on Control of Network Systems*, vol. 5, no. 4, pp. 1887-1897, Dec. 2018
21. Y. Zhou, D. K.Y. Yau, P. You, and P. Cheng. **Optimal-Cost Scheduling of Electrical Vehicle Charging under Uncertainty**. *IEEE Transactions on Smart Grid*, vol. 9, no. 5, pp. 4547-4554, Sep. 2018
22. H. Ma, Z. Yang, P. You, and M. Fei. **Multi-objective Biogeography-based Optimization for Dynamic Economic Emission Load Dispatch Considering Plug-in Electric Vehicles Charging**. *Energy*, vol. 135, pp. 101-111, Sep. 2017
23. P. You, Z. Yang, Y. Zhang, S. H. Low, and Y. Sun. **Optimal Charging Schedule for a Battery Switching Station Serving Electric Buses**. *IEEE Transactions on Power Systems*, vol. 31, no. 5, pp. 3473-3483, Sep. 2016
24. P. You, Z. Yang, M.-Y. Chow, and Y. Sun. **Optimal Cooperative Charging Strategy for a Smart Charging Station of Electric Vehicles**. *IEEE Transactions on Power Systems*, vol. 31, no. 4, pp. 2946-2956, Jul. 2016

25. Z. Yang, R. Wu, J. Yang, K. Long, and P. You. **Economical Operation of Microgrid with Various Devices via Distributed Optimization**. *IEEE Transactions on Smart Grid*, vol. 7, no. 2, pp. 857-867, Mar. 2016
26. Z. Yang, K. Long, P. You, and M.-Y. Chow. **Joint Scheduling of Large-Scale Appliances and Batteries via Distributed Mixed Optimization**. *IEEE Transactions on Power Systems*, vol. 30, no. 4, pp. 2031-2040, Jul. 2015

Conference proceedings

1. Y. Yu, R. K. Bansal, Y. Jiang, and P. You. **Learning-Augmented Primal-Dual Control Design for Secondary Frequency Regulation**. *IEEE American Control Conference* (*ACC*), New Orleans, US, 2026
2. Y. Liu, Y. Liu, and P. You. **Coherency Analysis in Nonlinear Heterogeneous Power Networks: A Blended Dynamics Approach**. *ACM e-Energy*, Banff, Canada, 2026
3. C. Qu, H. Jia, and P. You. **Decision-Dependent Distributionally Robust Optimization with Application to Dynamic Pricing**. *IEEE Conference on Decision and Control* (*CDC*), Rio de Janeiro, Brazil, 2025
4. K. Mukhi, C. Qu, P. You, and A. Abate. **Distributionally Robust Aggregation of Electric Vehicle Flexibility**. *ACM International Conference on Hybrid Systems: Computation and Control* (*HSCC*), Irvine, USA, 2025
5. R. Liu, Y. Pan, L. Xu, L. Song, P. You, Y. Chen, and J. Bian. **Efficient Discovery of Pareto Front for Multi-Objective Reinforcement Learning**. *International Conference on Learning Representations* (*ICLR*), Singapore, 2025
6. C. Qu, L. Shi, K. Panaganti, P. You, and A. Wierman. **Hybrid Transfer Reinforcement Learning: Provable Sample Efficiency From Shifted-Dynamics Data**. *International Conference on Artificial Intelligence and Statistics* (*AISTATS*), Phuket, Thailand, 2025 (**Oral, 2%**)
7. Y. Liu, E. Mallada, Z. Li, and P. You. **Accelerated Saddle Flow Dynamics for Bilinearly Coupled Minimax Problems**. *IEEE Conference on Decision and Control* (*CDC*), Milan, Italy, 2024
8. H. Xu, E. Liu, and P. You. **Statistically Feasible Mixed-Integer Programming for Integrated Energy System Scheduling**. *IEEE Conference on Energy Internet and Energy System Integration*, Shenyang, China, 2024
9. Y. Qiu, P. You, and Z. Zhao. **Reserve or Not? Ordering Emergency Supplies from an Opportunistic Supplier Facing Yield and Demand Uncertainties**. *INFORMS MSOM Conference*, Twin Cities, US, 2024
10. K. Gao, Y. Yan, Y. Liu, E. Liu, and P. You. **Online Electricity Purchase for Data Center with Dynamic Virtual Battery from Flexibility Aggregation**. *Youth Academic Annual Conference of Chinese Association of Automation* (*YAC*), Dalian, China, 2024
11. H. Zhao, Y. Yan, E. Liu, and P. You. **A Data-Driven Simulator for Flexible Electric Vehicle Charging: Framework, Model, and Algorithm**. *International Conference on Automation Science and Engineering* (*CASE*), Auckland, New Zealand, 2023
12. T. Zheng, P. You, and E. Mallada. **Constrained Reinforcement Learning via Dissipative Saddle Flow Dynamics**. *Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, US, 2022
13. R. K. Bansal, Y. Chen, P. You, and E. Mallada. **Intercept Bidding in Two-stage Settlement Electricity Markets**. *ACM e-Energy*, Virtual Event, 2022
14. R. K. Bansal, P. You, D. F. Gayme, and E. Mallada. **A Market Mechanism for Truthful Bidding with Energy Storage**. *Power Systems Computation Conference* (*PSCC*), Porto, Portugal, 2022
15. K. Lin, B. Wang, and P. You. **Mechanism Design for Efficient Nash Equilibrium in Oligopolistic Markets**. *Hawaii International Conference on System Sciences* (*HICSS*), Maui Island, US, 2022
16. P. You, and E. Mallada. **Saddle Flow Dynamics: Observable Certificates and Separable Regularization**. *American Control Conference* (*ACC*), New Orleans, US, 2021
17. R. K. Bansal, P. You, D. F. Gayme, and E. Mallada. **Storage Degradation Aware Economic Dispatch**.

American Control Conference (*ACC*), New Orleans, US, 2021

18. P. You, D. F. Gayme, and E. Mallada. **The Role of Strategic Load Participants in Two-Stage Settlement Electricity Markets**. *IEEE Conference on Decision and Control* (*CDC*), Nice, France, 2019
19. H. Fan, P. You, Z. Guo, S. Chen, and Z. Yang. **Online Pricing Mechanism for Electric Vehicles Charging Based on Operational Condition of A Charging Station**. *Chinese Control Conference* (*CCC*), Guangzhou, China, 2019
20. H. Fan, S. Chen, Z. Guo, P. You, and Z. Yang. **Coordinated Scheduling Strategy of Charging Station Considering Cost and Efficiency**. *IEEE International Conference on Control & Automation* (*ICCA*), Edinburgh, Scotland, 2019
21. P. You*, J. Z.F. Pang*, and E. Yeung*. **Deep Koopman Controller Synthesis for Cyber-Resilient Market-Based Frequency Regulation** (*all authors contributed equally). *IFAC Symposium on Control of Power and Energy Systems* (*CPES*), Tokyo, Japan, 2018
22. P. You, J. Z.F. Pang, and E. Yeung. **Stabilization of Power Networks via Market Dynamics**. *ACM e-Energy*, Karlsruhe, Germany, 2018
23. P. You, J. Z.F. Pang, S. H. Low, and P. Cheng. **Efficient Online Station Assignment for EV Battery Swapping** (poster abstract). *ACM e-Energy*, Karlsruhe, Germany, 2018
24. X. Liu, P. You, and Z. Yang. **Bilaterally Open Electricity Markets with High Penetration of Renewables**. *IEEE International Conference on Control & Automation* (*ICCA*), Anchorage, US, 2018
25. Z. Yu, P. You, and Z. Yang. **Residential Load Reduction Scheduling with Optimal Power Flow and Segmented Incentives**. *IEEE PES General Meeting*, Portland, US, 2018
26. J. Z.F. Pang, P. You, and M. Chen. **The Role of Information in Temporally Networked Cournot Markets: Second Chance in Power Markets**. *Hawaii International Conference on System Sciences* (*HICSS*), Hawaii, US, 2018
27. P. You, J. Z.F. Pang, M. Chen, S. H. Low, and Y. Sun. **Battery Swapping Assignment for Electric Vehicles: A Bipartite Matching Approach**. *IEEE Conference on Decision and Control* (*CDC*), Melbourne, Australia, 2017
28. H. Ma, P. You, K. Liu, Z. Yang, and M. Fei. **Optimal Battery Charging Strategy Based on Complex System Optimization**. *International Conference on Intelligent Computing for Sustainable Energy and Environment* (*ICSEE*), Nanjing, China, 2017
29. P. You, Y. Sun, J. Z.F. Pang, S. H. Low, and M. Chen. **Battery Swapping Assignment for Electric Vehicles: A Bipartite Matching Approach** (extended abstract). *Greenmetrics*, Urbana-Champaign, US, 2017
30. T. Guo, P. You, and Z. Yang. **Recommendation of Geo-Distributed Charging Stations for EVs: A Game Theoretical Approach**. *IEEE PES General Meeting*, Chicago, US, 2017
31. H. Ma, Z. Yang, P. You, and M. Fei. **Complex System Optimization for Economic Emission Load Dispatch**. *International Conference on Control*, Belfast, Northern Ireland, UK, 2016
32. E. Liu, P. You, and P. Cheng. **Optimal Privacy-Preserving Load Scheduling in Smart Grid**. *IEEE PES General Meeting*, Boston, US, 2016
33. P. You, S. H. Low, Z. Yang, Y. Zhang, and L. Fu. **Real-Time Recommendation Algorithm of Battery Swapping Stations for Electric Taxis**. *IEEE PES General Meeting*, Boston, US, 2016
34. L. Mo, P. You, X. Cao, Y.-Q. Song, and J. Chen. **Decentralized Multi-Charger Coordination for Wireless Rechargeable Sensor Networks**. *International Performance Computing and Communications Conference* (*IPCCC*), Nanjing, China, 2015
35. P. You, Z. Yang, and Y. Zhang. **Efficient Battery Charging Schedule of Battery-Swapping Station for Electric Buses**. *European Control Conference* (*ECC*), Linz, Austria, 2015
36. P. You, and Z. Yang. **Efficient Optimal Scheduling of Charging Station with Multiple Electric Vehicles via V2V**. *IEEE SmartGridComm*, Venice, Italy, 2014

PATENTS

Resource Configuration Methodology and System for Cloud Service Management Platform
China, granted in 2024

MENTORSHIP

Advisor

Yang Hu **Peking University**
Master student since 2025

Yixuan Yu **Peking University**
Ph.D. student since 2024

Yixuan Liu **Peking University**
Ph.D. student since 2024

Hengxing Zhao **Peking University**
Ph.D. student since 2023

Shufan Yu **Peking University**
Master student since 2023

Hanjiang Lin **Peking University**
Master student since 2023

Zijian Zhao **Peking University**
Ph.D. student since 2022, co-advised with Yunzhe Qiu

Yuke Zhou **Peking University**
Ph.D. student since 2022, co-advised with Jie Song

Yingzhu Liu **Peking University**
Ph.D. student since 2022, co-advised with Zhongkui Li

Huilan Xu **Peking University**
M.Eng, Sep. 2022 - Jun. 2025 *Next hop: China Southern Power Grid*
Thesis: Data-Driven Robust Scheduling of Integrated Energy System

Kekun Gao **Peking University**
M.Eng, Sep. 2022 - Jun. 2025 *Next hop: General Office of Shandong Provincial People's Government*
Thesis: Multi Timescale Electricity Trading Decision-Making for Low-Carbon Data Centers

Co-mentor

Rajni Kant Bansal **Johns Hopkins University**
Ph.D. 2023 *Next hop: Postdoc @UCSD*
Thesis: Efficiency and Market Power in Electricity Markets with Inelastic Demand, Energy Storage, and Hybrid Energy Resources

Xianglu Liu **Zhejiang University**
M.Sc. 2019 *Next hop: State Grid*
Thesis: Reformed Electricity Market with Deep Penetration of Renewables

Ziwen Yu **Zhejiang University**
M.Sc. 2018 *Next hop: Hikvision*
Thesis: Experimental Testbed and Optimal Load Scheduling for Demand Response Management in Smart Grid

Tianci Guo **Zhejiang University**
M.Sc. 2017 *Next hop: Huawei*
Thesis: Joint Temporal-Spatial Coordination of Charging Decision of Plug-in Electric Taxi Fleet

Host

Jiayi Li — **Peking University**
Undergraduate since Sep. 2025

Zhuo Ouyang — **Peking University**
Undergraduate since Jul. 2024

Chengrui Qu — **Peking University**
Undergraduate, Mar. 2023 - Jul. 2025 — *Next hop: Ph.D. student at Caltech*
Thesis: Hybrid Transfer Reinforcement Learning: Provable Sample Efficiency From Shifted-Dynamics Data

Yixuan Wang — **Peking University**
Undergraduate, Jun. 2023 - Aug. 2024 — *Next hop: Master student at Peking University*

Endong Liu — **Heze College**
Visiting scholar, Aug. 2022 - Jul. 2024

Yi Yan — **Peking University**
Undergraduate, Sep. 2022 - Jun. 2024 — *Next hop: Master student at National University of Singapore*

Xiwen Huang — **Peking University**
Research assistant, Aug. 2022 - Jul. 2023 — *Next hop: Ph.D. student at Imperial College London*

Hengxing Zhao — **Peking University**
Undergraduate, May 2022 - Jun. 2023 — *Next hop: Ph.D. student at Peking University*

Fuyi Li — **Peking University**
Undergraduate, Jul. 2022 - Feb. 2023 — *Next hop: Ph.D. student at Peking University*

Ruishuang Chen — **Southern University of Science and Technology**
Visiting student, Aug. 2022 - Nov. 2022

ACADEMIC ACTIVITIES

Community service

- Technical committee
  1. TC on Embodied AI, Chinese Association of Automation — *2024 - Present*
  2. TC on Energy Systems, IEEE Control Systems Society — *2023 - Present*
- Summer school
  1. 1st Graduate Summer School on Data and OR Intelligence @Peking University — *Jul. 2025*

Editorial board

- Guest editor
  1. Frontiers in Energy Research — *2022*
     Special Issue on AI, Data Analytics, and Mechanism Design for DER Integration Toward Net Zero

Conference organization

- Session chair
  1. Conference on Fully Actuated System Theory and Applications — *May 2024*
     Towards Next-Gen Power and Energy Systems
  2. POMS International Conference in China — *Jun. 2022*
     Data-Driven Operations Management
  3. SIAM Conference on Control and Its Applications — *Jul. 2021*
     Analysis and Control of Nonlinear Systems
  4. Conference on Information Sciences and Systems — *Mar. 2021*
     Power System Control, Optimization and Economics I & II
     Systems and Control

5. Conference on Information Sciences and Systems *Mar. 2019*
   Security and Privacy

- TPC member
  1. PowerUp *Sep. 2026*
  2. ACM e-Energy *Jun. 2026*
  3. IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids *Nov. 2023*
  4. IEEE Global Communications Conference *Dec. 2023*
  5. IEEE Global Communications Conference *Dec. 2022*
  6. IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids *Oct. 2022*
  7. IEEE Global Communications Conference *Dec. 2021*
  8. Conference on Information Sciences and Systems *Mar. 2021*
  9. International Workshop on Cyberspace Security *Dec. 2020*

Dissertation Committee

- Xinming Wang, Peking University *May 2025*
  Transfer Learning Methods based on Gaussian Process
- Xinming Han, Peking University *May 2025*
  Cloud Computing Resource Scheduling based on Robust Optimization
- Chenbei Lu, Tsinghua University *May 2025*
  Sample-Based Decision-Making Framework for Uncertainty-Aware Power System Operations
- Ruiyang Jin, Peking University *Jun. 2024*
  Modeling and Optimization of Multi-Agent Systems on Demand Side in Power Systems
- Jiayuan Wu, Peking University *Jun. 2024*
  Stuctured Non-Convex Optimization Theory and Algorithms
- Nan Gu, Tsinghua University *May 2024*
  Effective Decision-Making for an Active Power Distribution Network: Uncertainty Management and Mechanism Design
- Rajni K. Bansal, Johns Hopkins University *Aug. 2023*
  Efficiency and Market Power in Electricity Markets with Inelastic Demand, Energy Storage, and Hybrid Energy Resources
- Yongkang Ding, Peking University *May 2023*
  Deep Reinforcement Learning-Based Optimization Control Method for Mobile Energy Storage Systems
- Jiasheng Zhang, Tsinghua University *May 2023*
  Empowerment Mechanism Design for Power System Flexibility in the Context of Carbon Neutrality
- Jingshi Cui, Tsinghua University *May 2023*
  Operation Framework Evolution Research for Electricity Balancing Market Based on Market Entity's Participating Modes
- Fan Zhang, Peking University *May 2022*
  Decision-Making Optimization of Data-Driven Complex Stochastic Systems

Peer review

- Journal: Nature Communications, TAC, TCNS, TCST, TPS, TSG, TSE, TITS, TIE, TII, TPDS, TECS, Automatica, IEEE L-CSS, IEEE Systems Journal, IEEE Network, etc.
- Conference: ACM e-Energy, ACC, CDC, PSCC, CISS, PESGM, SmartGridComm, Globecom, etc.

Short visits

- California Institute of Technology, Dept. of Computing and Mathematical Sciences, Pasadena, California, US, Jun. 2023
- University of California San Diego, Dept. of Electrical and Computer Engineering, San Diego,

California, US, Jun. 2023
- Microsoft Research Asia, Machine Learning Group, Beijing, China, Apr. 2023 - Jul. 2023
- Southern University of Science and Technology, Dept. of Mechanical and Energy Engineering, Shenzhen, China, Jul. 2018 - Sep. 2018
- Chinese University of Hong Kong, Dept. of Information Engineering, Hong Kong, China, May 2018
- Chinese University of Hong Kong-Shenzhen, School of Science and Engineering, Shenzhen, China, Sep. 2016
- Singapore University of Technology and Design, Engineering Product Development Pillar, Singapore, Nov. 2015 - Feb. 2016

TALKS

Conference and workshop presentations

- "AI for Power System Operation", Forum on Power System Big Data Analytics, Nanjing University of Information Science & Technology, Nanjing, China, Sep. 2025
- "Hybrid Transfer Reinforcement Learning: Provable Sample Efficiency from Shifted-Dynamics Data", INFORMS International Conference, Singapore, Jul. 2025
- "Hybrid Transfer Reinforcement Learning: Provable Sample Efficiency from Shifted-Dynamics Data", International Workshop on Green Control, School of Science and Engineering, The Chinese University of Hong Kong - Shenzhen, Shenzhen, China, Mar. 2025
- "Mechanism Designs for Efficiency and Incentive Compatibility in Electricity Markets: Emerging Technology and Strategic Participation", Workshop on Power and Energy Systems of the (near) Future, Institute of High Performance Computing, ASTAR, Singapore, Dec. 2023
- "Saddle Flow Dynamics: Convergence, Algorithms, and Application to Constrained Reinforcement Learning", Young Scholar Workshop, Department of Automation, Shanghai Jiao Tong University, Shanghai, China, Mar. 2023
- "Saddle Flow Dynamics: Convergence, Algorithms, and Applications", MSRA StarTrack Program Workshop, Microsoft Research Asia, Beijing, China, Nov. 2022
- "Saddle Flow Dynamics: Convergence, Algorithms, and Applications", PKU Workshop on Operations Research and Machine Learning, Peking University, Beijing, China, Nov. 2022
- "Bidding Mechanisms and Incentive Analysis for Temporally-Coupled Electricity Markets", INFORMS Annual Meeting, Anaheim, California, US, Oct. 2021
- "Saddle Flow Dynamics: Observable Certificates and Separable Regularization", SIAM Conference on Control and Its Applications (CT), Spokane, Washington, US, July 2021
- "Saddle Flow Dynamics: Observable Certificates and Separable Regularization", IEEE American Control Conference (ACC), New Orleans, Louisiana, US, May 2021
- "The Role of Strategic Participants in Two-stage Settlement Electricity Markets", INFORMS Annual Meeting, Washington D.C., US, Nov. 2020
- "On Stability and Economic Efficiency of Electricity Market Dynamics", INFORMS Annual Meeting, Washington D.C., US, Nov. 2020
- "The Role of Strategic Load Participants in Two-Stage Settlement Electricity Markets", IEEE Conference on Decision and Control (CDC), Nice, France, Dec. 2019
- "The Role of Strategic Load Participants in Two-Stage Settlement Electricity Markets", Trans-Atlantic Infraday Conference, Washington, D.C., US, Oct. 2019
- "Deep Koopman Controller Synthesis for Cyber-Resilient Market-Based Frequency Regulation" (E. Yeung presented), IFAC Symposium on Control of Power and Energy Systems (CPES), Tokyo, Japan, Sep. 2018
- "Stabilization of Power Networks via Market Dynamics", ACM e-Energy, Karlsruhe, Germany, Jun. 2018

- "Online Station Assignment for EV Battery Swapping" (J. Z.F. Pang presented), IEEE Conference on Decision and Control (CDC), Melbourne, Australia, Dec. 2017
- "Online Station Assignment for EV Battery Swapping", Greenmetrics, Urbana-Champaign, Illinois, US, Jun. 2017
- "Optimal Charging Schedule for a Battery Switching Station Serving Electric Buses", IEEE PES General Meeting, Boston, Massachusetts, US, Jul. 2016
- "Joint Scheduling of Large-Scale Appliances and Batteries via Distributed Mixed Optimization", IEEE PES General Meeting, Denver, Colorado, US, Jul. 2015
- "Efficient Battery Charging Schedule of Battery-Swapping Station for Electric Buses", European Control Conference (ECC), Linz, Austria, Jul. 2015
- "Efficient Optimal Scheduling of Charging Station with Multiple Electric Vehicles via V2V", IEEE SmartGridComm, Venice, Italy, Nov. 2014

Seminar presentations

- "AI for Power and Energy Systems", International Joint Seminar, Yonsei University, Seoul, Korea, Jan. 2026
- "Mechanism Designs for Efficiency and Incentive Compatibility in Electricity Markets: Emerging Technology and Strategic Participation", School of Science and Engineering, The Chinese University of Hong Kong - Shenzhen, Shenzhen, China, Jan. 2024
- "Saddle Flow Dynamics: Convergence, Algorithms, and Applications", School of Data Science, The Chinese University of Hong Kong - Shenzhen, Shenzhen, China, Jan. 2024
- "Mechanism Designs for Efficiency and Incentive Compatibility in Electricity Markets: Emerging Technology and Strategic Participation", School of System Design and Intelligent Manufacturing, Southern University of Science and Technology, Shenzhen, China, Jan. 2024
- "Mechanism Designs for Efficiency and Incentive Compatibility in Electricity Markets: Emerging Technology and Strategic Participation", College of Electrical Engineering, Zhejiang University, Hangzhou, China, Jan. 2024
- "Saddle Flow Dynamics: Convergence, Algorithms, and Applications", College of Control Science and Engineering, Zhejiang University, Hangzhou, China, Jan. 2024
- "Saddle Flow Dynamics: Convergence, Algorithms, and Applications", National Engineering Laboratory for Big Data Analysis and Applications, Peking University, Beijing, China, Jan. 2024
- "Exploiting Flexibility of Electric Vehicle Charging", Microsoft Research Asia, Beijing, China, Jul. 2023
- "Saddle Flow Dynamics: Convergence, Algorithms, and Application to Constrained Reinforcement Learning", Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, US, Jun. 2023
- "Saddle Flow Dynamics: Convergence, Algorithms, and Application to Constrained Reinforcement Learning", Department of Automation, Tsinghua University, Beijing, China, Apr. 2023
- "Saddle Flow Dynamics: Convergence, Algorithms, and Application to Constrained Reinforcement Learning", AI Thrust, Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China, Mar. 2023
- "Saddle Flow Dynamics: Convergence, Algorithms, and Application to Constrained Reinforcement Learning", Department of Mechanical and Energy Engineering, Southern University of Science and Technology, Shenzhen, China, Jan. 2023
- "Saddle Flow Dynamics: Convergence, Algorithms, and Application to Constrained Reinforcement Learning", School of Computer Engineering, Jiangsu Ocean University, Lianyungang, China, Dec. 2022
- "Incentive Analysis and Coordination Design for Multi-Timescale Markets", Energy Seminar, Department of Environmental Health and Engineering, Johns Hopkins University, Baltimore, Maryland, US, Sep. 2021

- "Incentive Analysis and Coordination Design for Multi-Timescale Markets", Tsinghua-Berkeley Shenzhen Institute, Shenzhen, China, Jul. 2021
- "Incentive Analysis and Coordination Design for Multi-Timescale Markets", Department of Automation, Shanghai Jiao Tong University, Shanghai, China, May 2021
- "Incentive Analysis and Coordination Design for Multi-Timescale Markets", Department of Industrial Engineering and Management, Peking University, Beijing, China, May 2021
- "Incentive Analysis and Coordination Design for Multi-Timescale Markets", College of Smart Energy, Shanghai Jiao Tong University, Shanghai, China, Apr. 2021
- "Incentive Analysis and Coordination Design for Multi-Timescale Markets", Center for Future Networks, BNU-HKBU United International College, Zhuhai, China, Apr. 2021
- "On Stability and Economic Efficiency of Electricity Market Dynamics", International Forum of Excellent Young Talents in Engineering Sciences, Peking University, Beijing, China, Feb. 2021
- "On Stability and Economic Efficiency of Electricity Market Dynamics", Global Scientist Interdisciplinary Online Forum, Southern University of Science and Technology, Shenzhen, China, Jan. 2021
- "On Stability and Economic Efficiency of Electricity Market Dynamics", Vision Forum for International Young Scholars, Beijing University of Aeronautics and Astronautics, Beijing, China, Jan. 2021
- "The Role of Strategic Load Participants in Two-Stage Settlement Electricity Markets", School of Automation, Nanjing University of Science and Technology, Nanjing, China, Sep. 2019
- "The Role of Strategic Load Participants in Two-Stage Settlement Electricity Markets", School of Electrical Engineering, Southeast University, Nanjing, China, Sep. 2019
- "The Role of Strategic Load Participants in Two-Stage Settlement Electricity Markets", Department of Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China, Aug. 2019
- "The Role of Strategic Load Participants in Two-Stage Settlement Electricity Markets", Department of Mechanical and Energy Engineering, Southern University of Science and Technology, Shenzhen, China, Aug. 2019
- "The Role of Strategic Load Participants in Two-Stage Settlement Electricity Markets", Department of Control Science and Engineering, Zhejiang University, Hangzhou, China, Jul. 2019
- "The Role of Strategic Load Participants in Two-Stage Settlement Electricity Markets", Shanghai Astronomical Observatory, Chinese Academy of Science, Shanghai, China, Jul. 2019
- "The Role of Strategic Load Participants in Two-Stage Settlement Electricity Markets", Department of Computer Science, Central South University, Changsha, China, Jul. 2019
- "Scheduling of EV Battery Swapping", Department of Electrical Engineering, Tsinghua University, Beijing, China, Jul. 2018
- "Stabilization of Power Networks via Market Dynamics", Department of Information Engineering, Chinese University of Hong Kong, Hong Kong, China, May 2018
- "Scheduling of EV Battery Swapping", School of Science and Engineering, Chinese University of Hong Kong-Shenzhen, Shenzhen, China, Sep. 2016
- "Scheduling of EV Battery Swapping", SYSU-CMU Joint Institute of Engineering, Sun Yat-sen University, Guangzhou, China, Sep. 2016

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